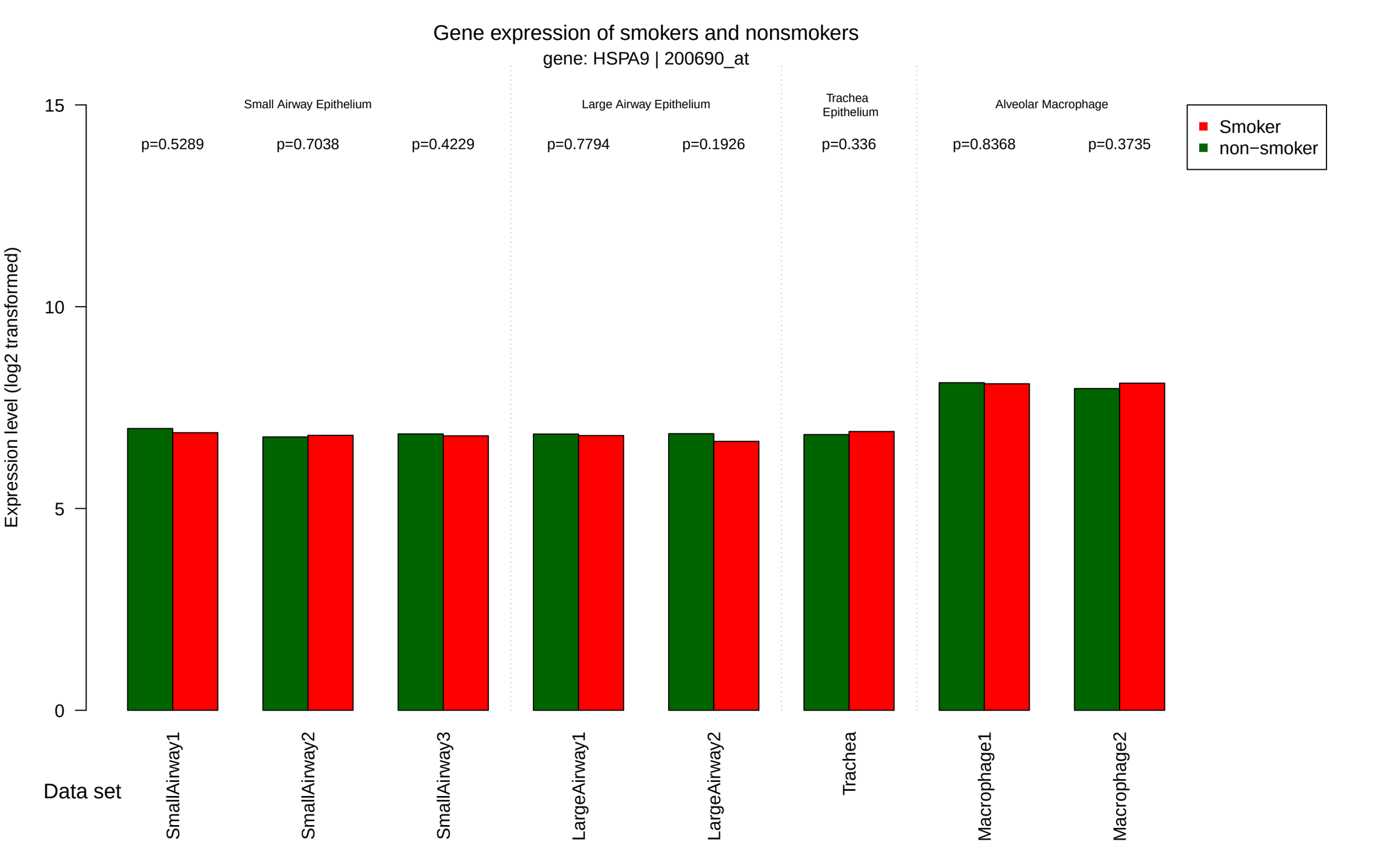
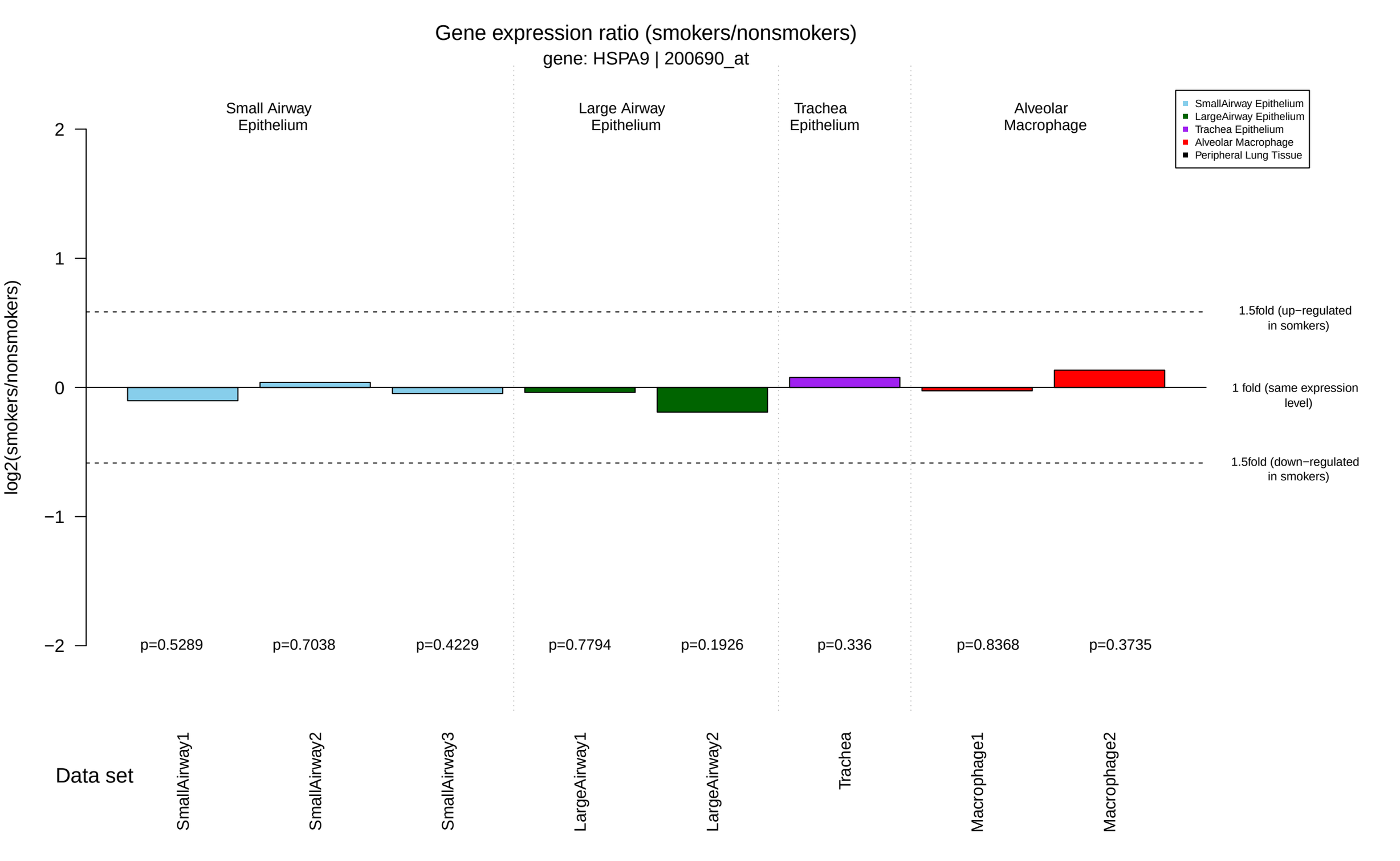
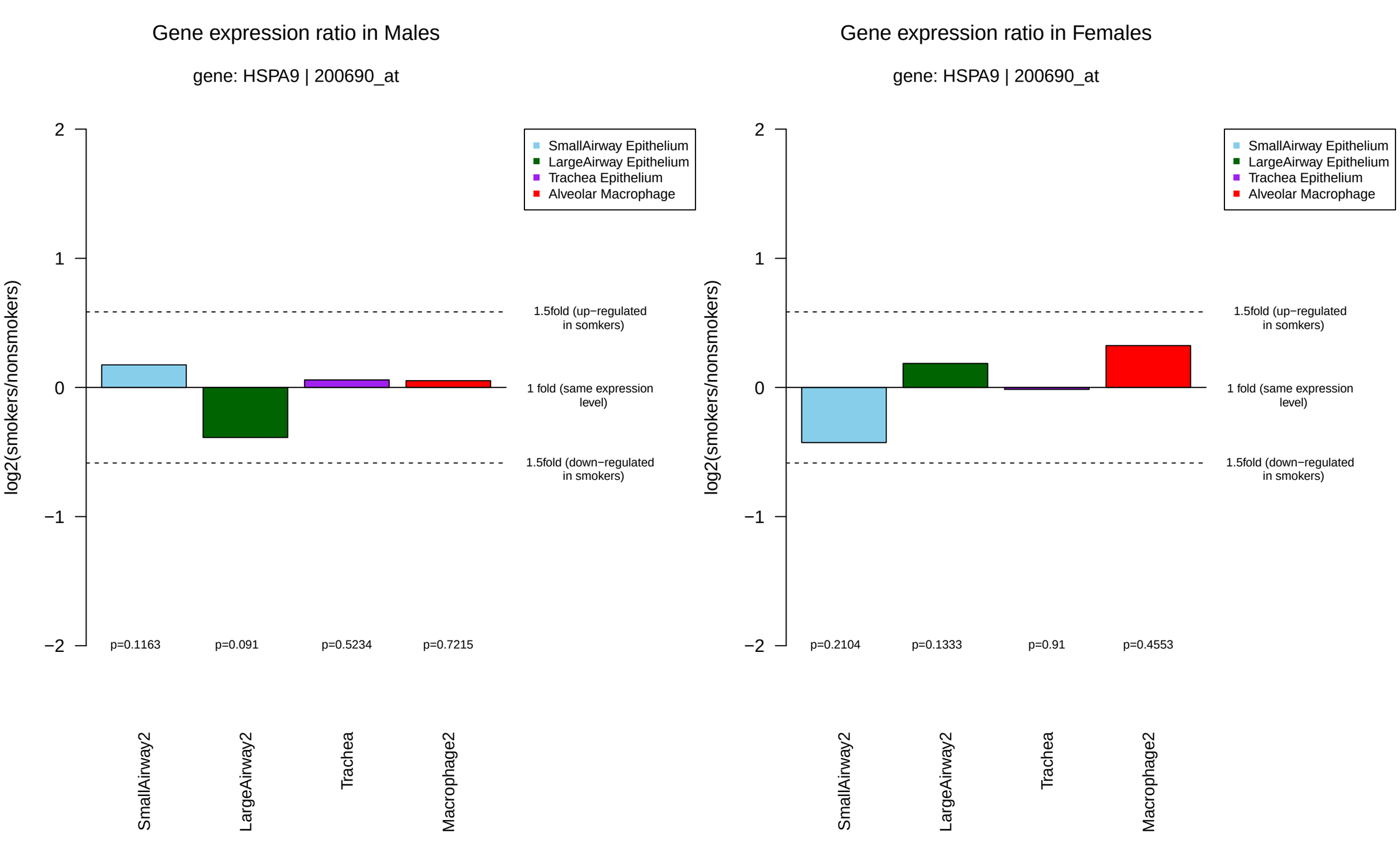
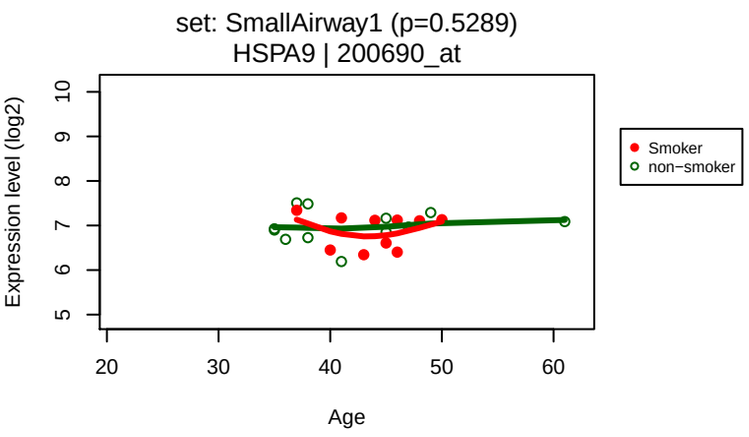
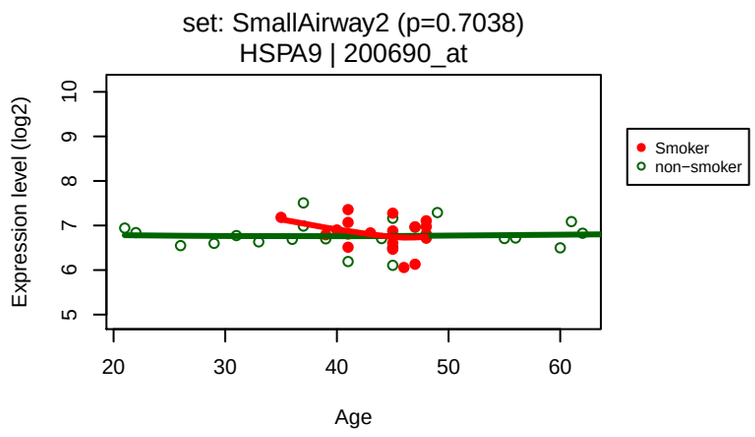
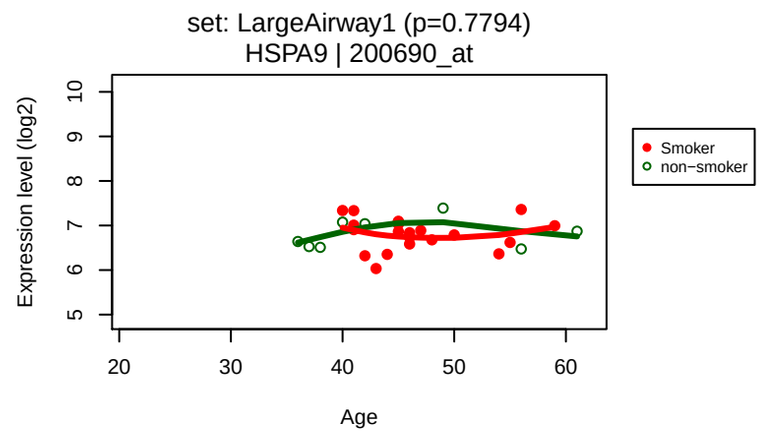

Gene expression of smokers and nonsmokers
gene: HSPA9 | 200690_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5289
p=0.7038
p=0.4229
p=0.7794
p=0.1926
p=0.336
p=0.8368
p=0.3735
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

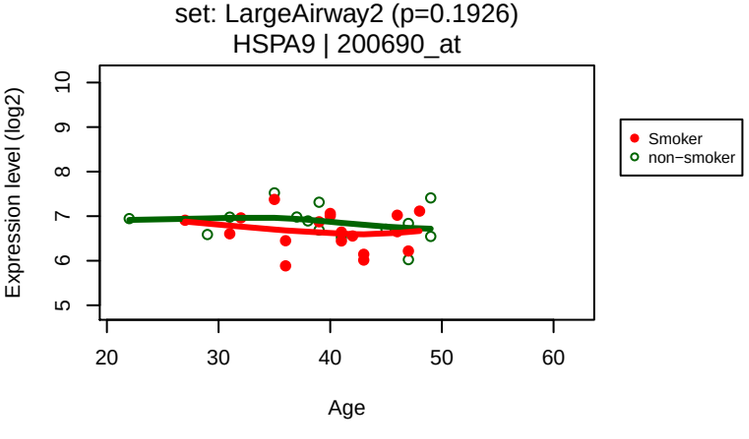

Gene expression ratio (smokers/nonsmokers)
gene: HSPA9 | 200690_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5289
p=0.7038
p=0.4229
p=0.7794
p=0.1926
p=0.336
p=0.8368
p=0.3735
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

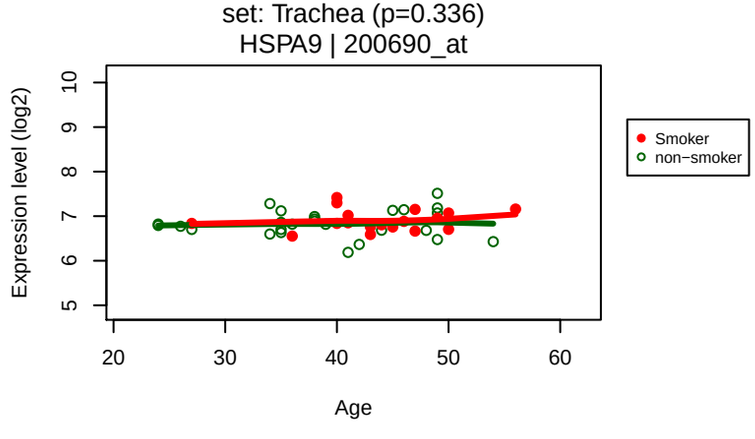

Gene expression ratio in Males
gene: HSPA9 | 200690_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1163
p=0.091
p=0.5234
p=0.7215
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HSPA9 | 200690_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2104
p=0.1333
p=0.91
p=0.4553
SmallAirway2
LargeAirway2
Trachea
Macrophage2

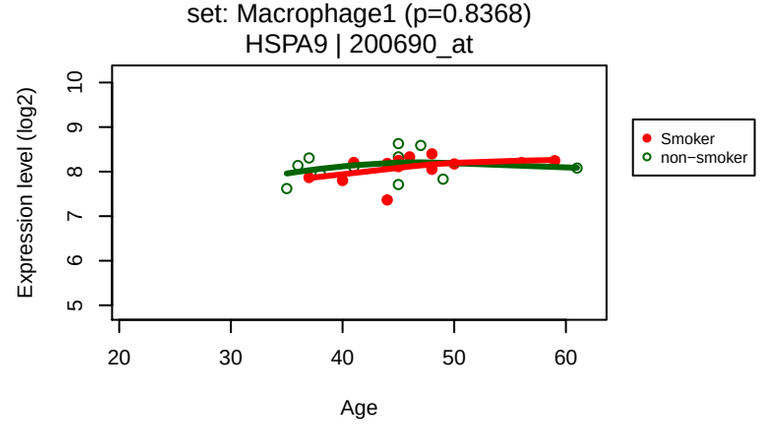

set: SmallAirway1 (p=0.5289)
HSPA9 | 200690_at
Expression level (log2)
Age
Smoker
non−smoker

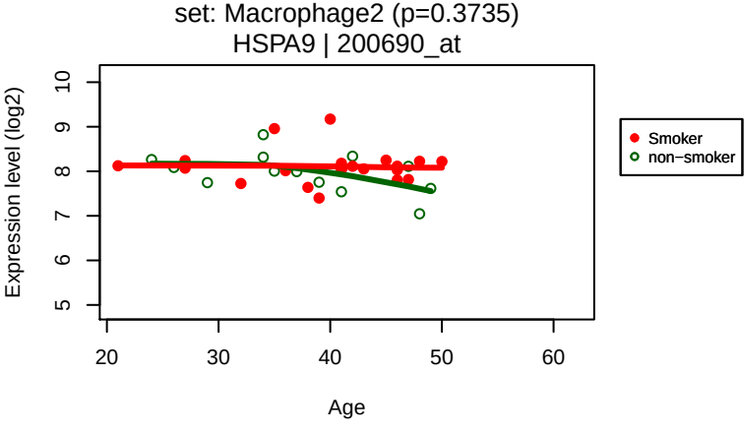

set: SmallAirway2 (p=0.7038)
HSPA9 | 200690_at
Expression level (log2)
Age
Smoker
non−smoker

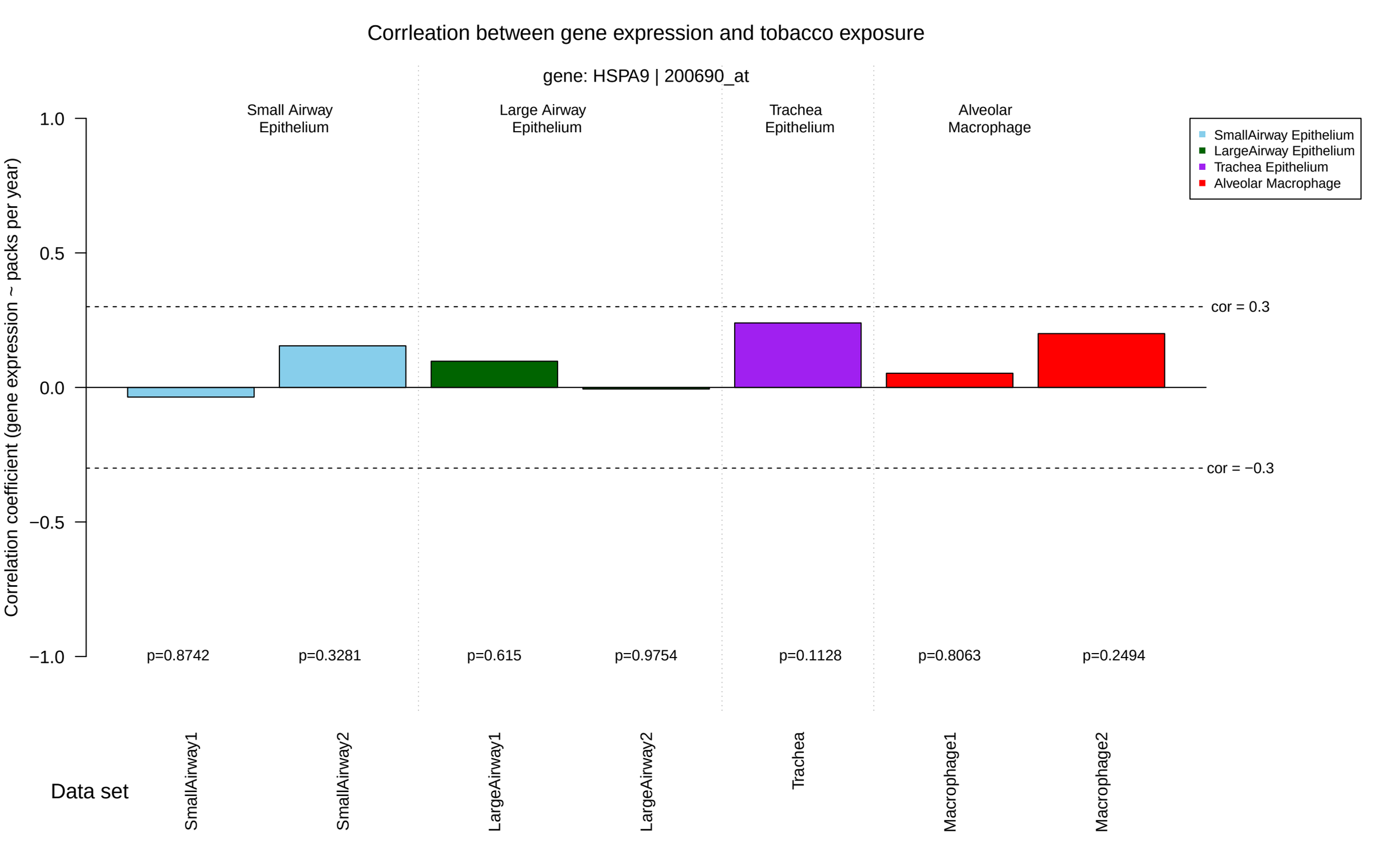

set: LargeAirway1 (p=0.7794)
HSPA9 | 200690_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1926)
HSPA9 | 200690_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.336)
HSPA9 | 200690_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8368)
HSPA9 | 200690_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3735)
HSPA9 | 200690_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HSPA9 | 200690_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8742
p=0.3281
p=0.615
p=0.9754
p=0.1128
p=0.8063
p=0.2494
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2